

NORMAL SUBGROUPS OF MAPPING CLASS GROUPS

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ABSTRACT. The mapping class group of a surface S , denoted $\mathrm{MCG}(S)$, is a topological group that captures the topological symmetries of S . In this survey, motivated by a result known as the Purity Theorem, we study normal subgroups of mapping class groups of different surfaces.

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1. PRELIMINARIES

We assume all surfaces to be second countable, connected, orientable, and with compact boundary.

1.1. Surfaces.

Let S be a surface. If $\pi_1(S)$ is finitely generated, we say that S is of *finite type*, and we say it is of *infinite type* otherwise.

Definition 1.1.1 (Space of ends). An exiting sequence in S is a sequence $\{U_n\}$ of connected open subsets of S such that

- (1) $m < n \Rightarrow U_n \subset U_m$.
- (2) U_n is not precompact for all n .
- (3) Any precompact subset of S is disjoint from all but finitely many U_n 's.

We say that two exiting sequences $\{U_n\}, \{V_n\}$ are equivalent if for any n , there exists N such that $U_N \subset V_n$ and vice versa.

We define that space of ends $\mathrm{Ends}(S)$, as the space of equivalence classes of exiting sequences of S , equipped with the following a topology:

Given a subset $U \subset S$ with compact boundary, we consider the set U^* of all ends represented by an exiting sequence eventually contained in U . The set

$$\{U^* : U \subset S \text{ open with compact boundary}\}$$

gives a basis for the topology of $\text{Ends}(S)$.

Theorem 1.1.2. For any surface S , the space $\text{Ends}(S)$ is totally disconnected, second countable, and compact. In particular, $\text{Ends}(S)$ is homeomorphic to a closed subset of a Cantor set.

Definition 1.1.3 (Planar and non-planar ends). We say that an end is *planar* if it admits a neighborhood that can be embedded in the plane. Otherwise, we say that it is *non-planar* or *accumulated by genus*, and every neighborhood of such end has infinite genus. We denote the (closed) subspace of non-planar ends as $\text{Ends}_{np}(S)$.

Theorem 1.1.4 (Classification of infinite-type surfaces). Let S_1, S_2 be surfaces, with genus g_1, g_2 , respectively, and b_1, b_2 boundary components, respectively. Then $S_1 \cong S_2$ if and only if $g_1 = g_2$, $b_1 = b_2$ and there is a homeomorphism

$$\text{Ends}(S_1) \rightarrow \text{Ends}(S_2)$$

that restricts to a homeomorphism $\text{Ends}_{np}(S_1) \rightarrow \text{Ends}_{np}(S_2)$.

For finite type surfaces, we let $S_{g,n}^b$ denote the surface of genus g with n punctures and b boundary components, and the omission of either b or n means that it is equal to 0.

1.2. Mapping Class Groups.

We now define the mapping class group and describe some of its elements.

Definition 1.2.1 (Mapping Class Groups). We define the *Mapping Class Group*

$$\text{MCG}(S) := \pi_0 \text{Homeo}^+(S, \partial S)$$

where $\text{Homeo}^+(S, \partial S)$ is equipped with the compact-open topology (which turns it into a topological group). We also define the *Extended Mapping Class Group*

$$\text{MCG}^\pm(S) := \pi_0 \text{Homeo}(S, \partial S).$$

Frequently, we will require a distinguished set $K \subset S$ to be fixed (not necessarily pointwise), and we denote the mapping class group of S relative to K as

$$\text{MCG}(S, K) := \pi_0 \text{Homeo}^+(S, \partial S; K),$$

where $\text{Homeo}^+(S, \partial S; K)$ is the subgroup of $\text{Homeo}^+(S, \partial S)$ that fixes K as a set.

Definition 1.2.2 (Pure Mapping Class Group). There is a natural homomorphism

$$\text{Homeo}^+(S, \partial S) \rightarrow \text{Homeo}(\text{Ends}(S), \text{Ends}_{np}(S)).$$

Since an isotopy fixes every end of a surface, this homomorphism factors through $\text{MCG}(S)$:

$$\text{MCG}(S) \rightarrow \text{Homeo}(\text{Ends}(S), \text{Ends}_{np}(S)).$$

This homomorphism is surjective, and its kernel is the *Pure Mapping Class Group* $\text{PMCG}(S)$, i.e. we have a short exact sequence

$$(1) \quad 1 \rightarrow \text{PMCG}(S) \rightarrow \text{MCG}(S) \rightarrow \text{Homeo}(\text{Ends}(S), \text{Ends}_{np}(S)) \rightarrow 1.$$

Note that when $S = S_{g,n}$, the above sequence reduces to

$$1 \rightarrow \text{PMCG}(S_{g,n}) \rightarrow \text{MCG}(S_{g,n}) \rightarrow \Sigma_n \rightarrow 1.$$

We now illustrate some elements of $\text{MCG}(S)$.

Definition 1.2.3 (Dehn twist). Let α be a simple closed curve in S , and let N_α be a tubular neighborhood of α . Let $\phi : S_{0,0}^2 \rightarrow N_\alpha$ be an orientation-preserving homeomorphism. We define a Dehn twist along α to be the isotopy class of $T_\alpha \in \text{Homeo}^+(S)$ given by

$$T_\alpha(x) = \begin{cases} \phi \circ T \circ \phi^{-1}, & x \in N_\alpha \\ x, & x \notin N_\alpha \end{cases}$$

where $T(r, \theta) = (r, \theta - 2\pi r)$, and the annulus $S_{0,0}^2$ is parametrized as $\{(r, \theta) \mid 1 \leq r \leq 2\}$ in polar coordinates.

Definition 1.2.4 (Hyperelliptic involution). We say that an element $\iota \in \text{MCG}(S)$ is a hyperelliptic involution if the induced automorphism of $H_1(S; \mathbb{Z})$ is $-\text{Id}$.

Definition 1.2.5 (Half twist). Let $S = S_{g,n}$ be a surface with marked points $\{x_1, \dots, x_n\}$, and consider a simple arc δ between x_i and x_{i+1} that is disjoint from all other x_j . Consider a regular neighborhood N_δ containing x_i, x_{i+1} that is disjoint from all other x_j . Let $\phi : D^2 \rightarrow N_\delta$ be an orientation preserving homeomorphism such that $\phi(1, \pi) = x_i$ and $\phi(1, 0) = x_{i+1}$, where D^2 is parametrized in polar coordinates as $\{(r, \theta) \mid 0 \leq r \leq 2\}$. We define a half twist along δ to be the isotopy class of $H_\delta \in \text{Homeo}(S)$ given by

$$H_\delta(x) = \begin{cases} \phi \circ H \circ \phi^{-1}, & x \in N_\delta \\ x, & x \notin N_\delta \end{cases}$$

where $H(r, \theta) = (r, \theta - \pi r)$.

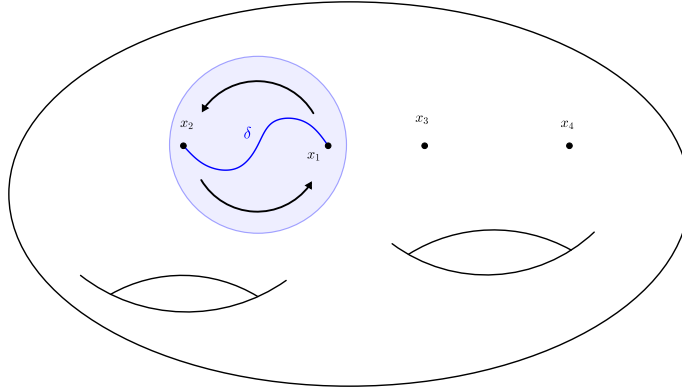


FIGURE 1. Half-twist between x_1 and x_2 along δ .

Definition 1.2.6 (Point-pushing). Let $S = S_{g,n}$ be a surface with marked points $\{x_1, \dots, x_n\}$, and let $[\gamma] \in \pi_1(S, x_i)$. Take an isotopy $f_t \in \text{Homeo}^+(S)$ such that $f_0 = \text{Id}$ and $f_t(x_i) = \gamma(t)$. We define point-pushing along $[\gamma]$ as $\mathcal{P}[\gamma] = f_1$.

Note that $\mathcal{P}[\gamma] \in \text{PMCG}(S)$, and that $\mathcal{P}$ gives a homomorphism $\pi_1(S, x_i) \rightarrow \text{MCG}(S)$.

2. THE BIRMAN EXACT SEQUENCE

In the finite type case, the half twists and point pushing maps turn out to be crucial for understanding a classical example of a normal subgroup of mapping class groups of punctured surfaces: the kernel of the forgetful map. This map arises by “forgetting” the distinguished set $K \subset S$ to give a homomorphism $\mathcal{F} : \text{MCG}(S, K) \rightarrow \text{MCG}(S)$.

2.1. Braid groups and mapping class groups.

Before describing the forgetful map more explicitly, we construct a group associated to a surface S with n marked points: the surface braid group $B_n(S)$. For most finite type surfaces, $B_n(S)$ embeds as a normal subgroup of $\text{MCG}(S, K)$ for $|K| = n$, and it serves as a geometric description of the kernel of the forgetful map.

Definition 2.1.1 (Configuration Spaces). Let S be a closed surface. We define the *ordered configuration space* of S as

$$\widehat{C}_n(S) = \left\{ x = (x_1, x_2, \dots, x_n) \in \prod_{i=1}^n S : x_i = x_j \Rightarrow i = j \right\},$$

endowed with the subspace topology of the product topology on $\prod_{i=1}^n S$. Note that the symmetric group Σ_n acts on $\widehat{C}_n(S)$ by permuting the entries of each x . We define the *configuration space* of S , $C_n(S)$ as the orbit space $C_n(S)/\Sigma_n$.

Definition 2.1.2 (Braid Groups). Let $x \in C_n(S)$. We define the *braid group* of S based at x as

$$B_n(S) := \pi_1(C_n(S), x),$$

and the *pure braid group*

$$PB_n(S) := \pi_1(\widehat{C}_n(S), x).$$

Braid groups and mapping class groups are closely related, thanks to the following proposition.

Proposition 2.1.3. Let S be a closed surface, and let $K = \{x_1, \dots, x_n\} \subset S$. The map $\mathcal{E}' : \text{Homeo}^+(S) \rightarrow \widehat{C}_n(S)$ given by

$$\mathcal{E}'(\phi) = (\phi(x_1), \dots, \phi(x_n))$$

gives a fiber bundle with fiber $\text{PHomeo}^+(S, K)$.

This fiber bundle yields the long exact sequence

$$\cdots \rightarrow \pi_i(\text{PHomeo}^+(S, K)) \rightarrow \pi_i(\text{Homeo}^+(S)) \rightarrow \pi_i(\widehat{C}_n(S)) \rightarrow \pi_{i-1}(\text{PHomeo}^+(S, K)) \rightarrow \cdots$$

The last few terms of the long exact sequence give

$$\cdots \rightarrow \pi_1(\text{Homeo}^+(S)) \rightarrow PB_n(S) \rightarrow \text{PMCG}(S, K) \rightarrow \text{MCG}(S) \rightarrow 1.$$

In particular, when $\chi(S) < 0$, we have that $\pi_1(\text{Homeo}^+(S)) = 1$, so we get a short exact sequence

$$1 \rightarrow PB_n(S) \rightarrow \text{PMCG}(S, K) \rightarrow \text{MCG}(S) \rightarrow 1.$$

Moreover, when $n = 1$ we have :

$$1 \xrightarrow{\mathcal{P}} \pi_1(S, x) \rightarrow \text{MCG}(S, x) \rightarrow \text{MCG}(S) \rightarrow 1,$$

where $\mathcal{P}$ is the point-pushing map from Definition 1.2.6.

Similarly, we obtain the following short exact sequence when $\chi(S) < 0$.

Theorem 2.1.4 (Birman Exact Sequence).

$$(2) \quad 1 \rightarrow B_n(S) \rightarrow \mathrm{MCG}(S, K) \xrightarrow{\mathcal{F}} \mathrm{MCG}(S) \rightarrow 1.$$

The map $\mathrm{MCG}(S, K) \xrightarrow{\mathcal{F}} \mathrm{MCG}(S)$ is known as the forgetful map, and the equivalence class $\mathcal{F}[\phi]$ of $\phi \in \mathrm{Homeo}^+(S, K)$ consists of all homeomorphisms of S that are isotopic to ϕ without restricting the isotopies to preserve the set K .

When (S, K) has finite type, $\ker \mathcal{F}$ is generated by half twists between the marked points x_i, x_{i+1} (which are pullbacks of transpositions in Σ_n under $\mathrm{MCG}(S_{g,n}) \rightarrow \Sigma_n$), and point-pushing maps along generators of $\pi_1(S, x_i)$.

When S has infinite type, or when the set K is infinite, we still get a surjection $\mathrm{MCG}(S, K) \xrightarrow{\mathcal{F}} \mathrm{MCG}(S)$, whose kernel may be harder to describe geometrically. For instance, when S has finite type and K is a Cantor set, Calegari and Chen prove that $\ker \mathcal{F}$ is generated by mapping classes supported in dividing discs, i.e. those that divide K into two homeomorphic copies of K , which is a key ingredient in their proof of the Purity Theorem; see Theorem 4.2.1 for more details.

3. NORMAL SUBGROUPS OF MAPPING CLASS GROUPS OF PUNCTURED SURFACES

So far, we have seen two important examples of normal subgroups of mapping class groups of punctured surfaces: $\mathrm{PMCG}(S)$, and $\ker \mathcal{F}$. In this section we illustrate other sources of normal subgroups of mapping class groups, with emphasis on non-compact surfaces of finite type.

3.1. Congruence subgroups.

Congruence subgroups exploit the algebraic topology of the surface S to construct normal subgroups of $\mathrm{MCG}(S)$. Since $\pi_1(S) \cong F_\infty$ for every infinite type surface, we assume S to have finite type for reasons that will become clear later. Note that every $f \in \mathrm{MCG}(S)$ induces an automorphism on $\pi_1(S)$ up to basepoints; more explicitly, there is a well-defined homomorphism $\mathrm{MCG}(S) \rightarrow \mathrm{Out}(\pi_1(S))$, which we can use to construct normal subgroups of $\mathrm{MCG}(S)$ via a smaller class of normal subgroups of $\pi_1(S)$, known as characteristic subgroups.

Before providing the definition of a congruence subgroup, we give a motivating construction of normal subgroups of mapping class groups coming from the homology of the surface. Note that any $f \in \mathrm{MCG}(S)$ induces an automorphism of $H_1(S; \mathbb{Z})$, so we have a homomorphism $\mathrm{MCG}(S) \rightarrow \mathrm{Sp}_{2g}(\mathbb{Z})$. With this in mind, we are ready to define principal congruence subgroups.

Definition 3.1.1 (Principal congruence subgroup for compact surfaces). We define the principal congruence subgroup of level m of $\mathrm{MCG}(S_g)$ as the kernel of the composition $\mathrm{MCG}(S_g) \rightarrow \mathrm{Sp}_{2g}(\mathbb{Z}) \rightarrow \mathrm{Sp}_{2g}(\mathbb{Z}/m)$, and we denote it by $\mathrm{MCG}(S_g)[m]$. The special case $\mathrm{MCG}(S_g)[0]$ is called the Torelli group, and we denote it by $\mathcal{I}(S_g)$.

The construction of the more general notion of congruence subgroups is motivated by a classical result known as the Dehn-Nielsen-Baer Theorem, which describes $\mathrm{MCG}(S)$ in terms of $\pi_1(S)$.

Theorem 3.1.2 (Dehn-Nielsen-Baer).

$$\mathrm{MCG}^\pm(S_g) \cong \mathrm{Out}(\pi_1(S_g))$$

We now exploit this isomorphism to construct normal subgroups of $\mathrm{MCG}(S_g)$.

Definition 3.1.3 (Characteristic subgroup). A subgroup $N \leq G$ is characteristic if it is invariant under every automorphism of G .

Definition 3.1.4 (Congruence subgroup for compact surfaces). Let $H \leq \pi_1(S_g)$ be a characteristic subgroup of finite index. We have a map

$$\mathrm{MCG}(S_g) \rightarrow \mathrm{Out}(\pi_1(S_g)/H).$$

Note that $\pi_1(S_g)/H$ is finite, hence $\mathrm{Out}(\pi_1(S_g)/H)$ is finite, so the kernel of the map $\mathrm{MCG}(S_g) \rightarrow \mathrm{Out}(\pi_1(S_g)/H)$ has finite index.

The kernel of any such homomorphism is called a congruence subgroup.

Remark 3.1.5. The map $\mathrm{MCG}(S_g) \rightarrow \mathrm{Out}(\pi_1(S_g)/H)$ is given as follows.

Write $G = \pi_1(S_g)$. We define a map $\mathrm{Aut}(G) \rightarrow \mathrm{Aut}(G/H)$ given by $\varphi \mapsto \bar{\varphi} = (xH \mapsto \varphi(x)H)$, which descends to a homomorphism $\mathrm{Out}(G) \rightarrow \mathrm{Out}(G/H)$. Indeed, let $\varphi_a \in \mathrm{Inn}(G)$ be conjugation by a , and let $x \in G$. Then $\bar{\varphi}_a(xH) = axa^{-1}H = axHa^{-1}$ (because H is normal), so $\bar{\varphi}_a \in \mathrm{Inn}(G/H)$. Lastly, we define $\mathrm{MCG}(S_g) \rightarrow \mathrm{Out}(\pi_1(S_g)/H)$ to be the restriction of the above homomorphism to $\mathrm{MCG}(S_g)$ as an index 2 subgroup of $\mathrm{Out}(\pi_1(S_g))$ by Dehn-Nielsen-Baer.

Remark 3.1.6. When H is the kernel of the composition $\pi_1(S_g) \rightarrow H_1(S_g; \mathbb{Z}) \rightarrow H_1(S_g; \mathbb{Z}/m)$ we obtain $\mathrm{MCG}(S_g)[m]$.

Example 3.1.7. Let A be any finite abelian group, and take H to be the kernel composition $\pi_1(S_g) \rightarrow H_1(S_g; \mathbb{Z}) \rightarrow H_1(S_g; A)$. Since A is finite, any such H will give rise to congruence subgroups of $\mathrm{MCG}(S_g)$ that contain $\mathcal{I}(S_g)$.

Example 3.1.8. Fix $k \geq 2$. Then the intersection of all index k subgroups of $\pi_1(S_g)$ is a characteristic subgroup. Indeed, any automorphism of $\pi_1(S_g)$ must permute all index k subgroups, hence it must fix their intersection. These need not contain $[\pi_1(S_g), \pi_1(S_g)]$, thus they give rise to congruence subgroups of $\mathrm{MCG}(S_g)$ that do not contain $\mathcal{I}(S_g)$.

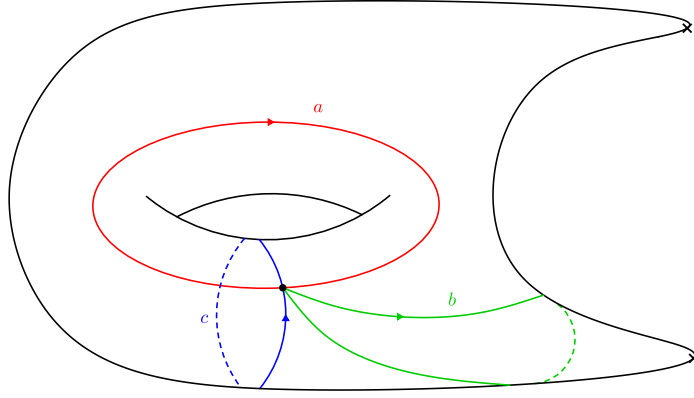
Question 1 (Congruence Subgroup Problem). Is it true that every finite-index subgroup in $\mathrm{MCG}(S_g)$ contains a congruence subgroup?

Remark 3.1.9. There is no immediate analogue of the Dehn-Nielsen-Baer for infinite type surfaces; this, along with the fact that $\pi_1(S) \cong F_\infty$ for any infinite type surface S , is an obstruction for defining congruence subgroups for infinite type surfaces in an analogous way.

Looking closer into the Dehn-Nielsen-Baer Theorem, the fact that $\mathrm{MCG}(S_g) \rightarrow \mathrm{Out}(\pi_1(S_g))$ is surjective reveals something crucial to be able to define congruence subgroups for punctured surfaces: every outer automorphism of $\pi_1(S_g)$ is induced by a homeomorphism of S_g . This fails to be true for punctured surfaces, as we will illustrate soon. Thus an analogue for Dehn-Nielsen-Baer will require us to restrict our attention to a smaller class of automorphisms of $\pi_1(S)$, namely those that are “geometric”.

Definition 3.1.10 (Geometric automorphism). An automorphism of $\pi_1(S)$ is said to be geometric if it is induced by a homeomorphism of S .

Example 3.1.11. Recall that $\pi_1(S) \cong F_3$. Consider the generating set $\{a, b, c\}$ for $\pi_1(S_{1,2})$ shown below.


 FIGURE 2. Generators a, b, c for $\pi_1(S_{1,2})$.

Note that any permutation of $\{a, b, c\}$ induces an automorphism of $\pi_1(S_{1,2})$, but any permutation with $a \mapsto b$ can't be realized by a homeomorphism of $S_{1,2}$, since a is non-separating, while b is separating.

Let $\text{Out}^*(\pi_1(S)) \leq \text{Out}(\pi_1(S))$ be the subgroup consisting of outer automorphisms that preserve the set of conjugacy classes of peripheral loops. The subgroup $\text{Out}^*(\pi_1(S))$ consists precisely of outer automorphisms coming from geometric automorphisms.

Theorem 3.1.12 (Dehn-Nielsen-Baer for punctured surfaces). Let $S = S_{g,n}$ be hyperbolic. Then

$$\text{MCG}^\pm(S) \cong \text{Out}^*(\pi_1(S)).$$

With this generalized version of the Dehn-Nielsen-Baer Theorem, the right analogue of a congruence subgroup for punctured surfaces is the following.

Definition 3.1.13. Suppose that $S = S_{g,n}$ for $n > 0$ is a punctured hyperbolic surface. Let $H \leq \pi_1(S)$ be a finite index geometrically characteristic subgroup, and consider the map

$$\phi_H : \text{Out}^*(\pi_1(S_{g,n})) \rightarrow \text{Out}(\pi_1(S_{g,n})/H).$$

Then we call $\ker \phi_H$ a congruence subgroup.

With this generalization, we again recover the definition of $\text{MCG}(S)[m]$ when S is a punctured surface by letting H be the kernel of the composition $\pi_1(S) \rightarrow H_1(S; \mathbb{Z}) \rightarrow H_1(S; \mathbb{Z}/m)$, which coincides with the kernel of the action on $H_1(S; \mathbb{Z}/m)$.

Turns out that, except for one case, principal congruence subgroups of level m of punctured surfaces of finite type are pure, as we show below.

Proposition 3.1.14. Let $g \geq 1$, and $m, n \geq 2$. Unless $n = m = 2$, then

$$\text{MCG}(S_{g,n})[m] \leq \text{PMCG}(S_{g,n}).$$

Proof. Given any $f \in \text{MCG}(S_{g,n})$, we let f_{*m} denote the induced automorphism of $H_1(S_{g,n}; \mathbb{Z}/m)$. We first treat the exceptional case $n = m = 2$. Consider the standard generators of $H_1(S_{g,2}; \mathbb{Z})$ $a_1, b_1, \dots, a_g, b_g, c_1$, where c_1 is a clockwise-oriented peripheral curve around one of the punctures, and let c_2 be a clockwise-oriented peripheral curve around the other puncture, so that $c_2 = -c_1$ in $H_1(S_{g,2}; \mathbb{Z})$.

Let Δ be a half twist between the two punctures, so $\Delta_{*0}(c_1) = c_2$. Since $c_2 = -c_1$ in $H_1(S_{g,2}; \mathbb{Z})$ and $-c_1 \equiv c_1 \pmod{2}$, then $\Delta_{*2}(a_i) = a_i$, $\Delta_{*2}(b_i) = b_i$, and $\Delta_{*2}(c_1) = c_1$, hence $\Delta \in \text{MCG}(S_{g,2})[2]$, and therefore $\text{MCG}(S_{g,2})[2] \not\leq \text{PMCG}(S_{g,2})$.

We now treat the case $(n, m) \neq (2, 2)$. Let $c_1, \dots, c_n$ be the homology classes of n distinct peripheral curves, oriented clockwise, and let $i \in \{1, \dots, n-1\}$. Since any $f \in \text{MCG}(S_{g,n})$ must send peripheral curves to peripheral curves, then there exists $j \in \{1, \dots, n\}$ such that $f_{*0}(c_i) = \pm c_j$. Now, assume that f acts trivially on $H_1(S_{g,n}; \mathbb{Z}/m)$. Since $f_{*0}(c_i) = \pm c_j$, then we must have that $f_{*0}(c_i) = \pm c_i$ for all $i \in \{1, \dots, n-1\}$. When $m = 2$ and $n \geq 3$, we may have that $f_{*0}(c_i) = -c_i$, but since $-c_i \neq \pm c_j$ for all j whenever $n \geq 3$, then $f \in \text{PMCG}(S_{g,n})$. Lastly, when $m \geq 3$, since $-c_i \not\equiv c_i \pmod{m}$, then in order for f to act trivially on $H_1(S_{g,n}; \mathbb{Z}/m)$, it must be that $f_{*0}(c_i) = c_i$ for all i , which similarly implies that $f \in \text{PMCG}(S_{g,n})$. $\blacksquare$

Remark 3.1.15. By the arguments above, $\Delta \notin \mathcal{I}(S_{g,2})$.

Remark 3.1.16. Even more, $\Delta_*^k(c_1) = (-1)^k c_1$ in $H_1(S_{g,2}; \mathbb{Z})$, so $\Delta^k \in \text{MCG}(S_{g,2})[m]$ for $m \geq 3$ if and only if $k \equiv 0 \pmod{2}$.

Observation 3.1.17. Since $\Delta \in \text{MCG}(S_{g,2})[2]$, then $\langle\langle \Delta \rangle\rangle \leq \text{MCG}(S_{g,2})[2]$, where $\langle\langle \Delta \rangle\rangle$ denotes the normal closure of Δ in $\text{MCG}(S_{g,2})$. In other words, any $f \in \langle\langle \Delta \rangle\rangle$ reduces to the trivial automorphism of $H_1(S_{g,2}; \mathbb{Z}/2)$.

Moreover, since point-pushing maps are not in $\text{MCG}(S_{g,2})[2]$, then

$$\langle\langle \Delta \rangle\rangle \neq \ker \mathcal{F}.$$

3.2. Pullbacks.

Recall that given a group homomorphism $\varphi : A \rightarrow B$, if $N \trianglelefteq B$, then $\varphi^{-1}(N) \trianglelefteq A$, hence a group homomorphism $\text{MCG}(S) \rightarrow G$ provides a normal subgroup of $\text{MCG}(S)$ for each normal subgroup of G .

Suppose that S is a closed surface and $K \subset S$ is a set of marked points in S . The short exact sequence (1) provides the homomorphism

$$\text{MCG}(S, K) \rightarrow \text{Homeo}(K)$$

and hence a source of normal subgroups of $\text{MCG}(S, K)$, namely those that contain $\text{PMCG}(S, K)$. Similarly, normal subgroups of $\text{MCG}(S)$ give rise to normal subgroups of $\text{MCG}(S, K)$ via the forgetful map

$$\text{MCG}(S, K) \xrightarrow{\mathcal{F}} \text{MCG}(S)$$

from (2), namely those that contain $\ker \mathcal{F}$.

Example 3.2.1. Let $\Gamma_m = \text{MCG}(S_g)[m] \trianglelefteq \text{MCG}(S_g)$. Then $\text{MCG}(S_{g,n})[m] \leq \mathcal{F}^{-1}(\Gamma_m)$ for all m , since any $f \in \text{MCG}(S_{g,n})$ that induces the identity map on $H_1(S_{g,n}; \mathbb{Z}/m) \cong (\mathbb{Z}/m)^{2g+n-1}$ also induces the identity map on $H_1(S_g; \mathbb{Z}/m) \cong (\mathbb{Z}/m)^{2g}$. However, for

$m, n \geq 3$, $\mathcal{F}^{-1}(\Gamma_m) \neq \text{MCG}(S_{g,n})[m]$. Indeed, let $\Delta \in \text{MCG}(S_{g,n})$ be a half-twist between two marked points. Note that $\Delta \in \ker \mathcal{F} \leq \mathcal{F}^{-1}(\Gamma_m)$, but by Proposition 3.1.14 $\Delta \notin \text{MCG}(S_{g,n})[m]$.

Example 3.2.2. From the homomorphism $\text{MCG}(S_{g,n}) \rightarrow \Sigma_n$, when $n \geq 5$, recall that the only non-trivial proper subgroup of Σ_n is the alternating group A_n . Hence for $n \geq 5$, the only normal subgroups of $\text{MCG}(S_{g,n})$ that pull back from Σ_n are $\text{PMCG}(S_{g,n})$, $\text{MCG}(S_{g,n})$ and the pullback of A_n .

Remark 3.2.3. When $\text{Homeo}(K)$ is simple (e.g. when $|K| = 2$ or when K is a Cantor set), the homomorphism $\text{MCG}(S, K) \rightarrow \text{Homeo}(K)$ yields only $\text{PMCG}(S, K)$ and $\text{MCG}(S, K)$ as normal subgroups.

3.3. Normal closures.

The final source of normal subgroups of $\text{MCG}(S)$ that we study is the normal closure of certain mapping classes.

Throughout this section, we assume $n \geq 3$ and $g \geq 1$. Let $x_1, \dots, x_n$ be the marked points of $S_{g,n}$, and let $\{x_1, \dots, x_n\} \subset D \hookrightarrow S_{g,n}$, where D is an embedded disc. We let $\Delta_1, \dots, \Delta_{n-1} \in \text{MCG}(S_{g,n})$ be half-twists between x_i and x_{i+1} , respectively, with $\text{supp}(\Delta_i) \subset D_i \subset D$, where D_i is an embedded disc containing x_i and x_{i+1} , but no other x_j .

Let $B_n(D)^{B_n(S_g)}$ denote the normal closure of $B_n(D)$ in $B_n(S_g)$. Since $B_n(D)$ is generated by half-twists, which are all conjugate, then the image of $B_n(D)^{B_n(S_g)}$ under the map $B_n(S_g) \rightarrow \text{MCG}(S_{g,n})$ from the Birman Exact Sequence is precisely $\langle\langle \Delta_1 \rangle\rangle$. From the presentation given in [6], the quotient of $B_n(S_g)$ by $B_n(D)^{B_n(S_g)}$ is isomorphic to $H_1(S_g; \mathbb{Z})$ (in particular, it's not trivial). Since $\ker \mathcal{F} \cong B_n(S_g)$ for $g \geq 2$, it follows that we can't find point-pushing maps in the normal closure of Δ_1 in $\ker \mathcal{F}$, since point-pushing maps along non-separating simple loops are conjugate.

Since $\langle\langle \Delta_1 \rangle\rangle_{\ker \mathcal{F}} \neq \ker \mathcal{F}$ by the observations above, we may ask: how much larger is $\langle\langle \Delta_1 \rangle\rangle$ compared to $\langle\langle \Delta_1 \rangle\rangle_{\ker \mathcal{F}}$? The following lemmas show that these normal closures are, in fact, equal.

Lemma 3.3.1. *Let $g \geq 1$. Then*

$$B_n(S_g)^{\text{ab}} \xrightarrow[q]{\cong} \mathbb{Z}^{2g} \times \mathbb{Z}/2.$$

Lemma 3.3.2. *Let $g \geq 1$, and let $\delta = (0, \dots, 0, 1) \in \mathbb{Z}^{2g} \times \mathbb{Z}/2$. Then*

$$B_n(D)^{B_n(S_g)} = q^{-1}\langle\delta\rangle.$$

When $g \geq 2$, the Birman Exact Sequence (2) provides an embedding $B_n(S_g) \hookrightarrow \text{MCG}(S_{g,n})$ by identifying $B_n(S_g)$ with $\ker \mathcal{F}$. We thus have the following chain of normal subgroups

$$q^{-1}\langle\delta\rangle \trianglelefteq \ker \mathcal{F} \trianglelefteq \text{MCG}(S_{g,n}).$$

Lemma 3.3.3. *Let $g \geq 2$, and identify $B_n(S_g) \cong \ker \mathcal{F} \trianglelefteq \text{MCG}(S_{g,n})$. Then $q^{-1}\langle\delta\rangle \trianglelefteq \text{MCG}(S_{g,n})$.*

For $g \geq 2$, by identifying $q^{-1}\langle\delta\rangle$ with the normal closure of Δ_1 in $\ker \mathcal{F}$, the above Lemma implies that $\langle\langle \Delta_1 \rangle\rangle = \langle\langle \Delta_1 \rangle\rangle_{\ker \mathcal{F}}$, which is properly contained in $\ker \mathcal{F}$. With a bit more work, this result can be extended to $S = T^2$ to conclude that $\langle\langle \Delta_1 \rangle\rangle = \langle\langle \Delta_1 \rangle\rangle_{\ker \mathcal{F}}$ regardless

of the genus of the finite type surface S .

We may also inquire about the normal closure of other mapping classes in $\text{MCG}(S_{g,n})$. For example, letting c be a non-separating simple closed curve, it is straightforward to show that $\langle\langle T_c \rangle\rangle = \text{MCG}(S_g)$. Naturally, we also ask: what other mapping classes normally generate $\text{MCG}(S_{g,n})$? In their paper [14], Margalit and Lanier develop a theory for normal generators of mapping class groups, which is slightly generalized in [4]. We now survey part of their theory.

Definition 3.3.4 (Graphs of curves). We let $\mathcal{N}(S_g)$ denote the graph of non-separating curves for S_g , and we let $\mathcal{N}_f(S_g)$ denote the abstract graph whose vertex set is the same as that of $\mathcal{N}(S_g)$, and whose edges correspond to pairs of vertices $\{c, h(c)\}$, where h is conjugate to f .

Proposition 3.3.5 (General well-suited curve criterion). Let $f \in \text{MCG}(S_{g,n})$. If $\mathcal{N}_f(S_{g,n})$ is connected, then

$$[\text{PMCG}(S_{g,n}), \text{PMCG}(S_{g,n})] \leq \langle\langle f \rangle\rangle.$$

When $g \geq 3$, $\mathcal{N}_f(S_{g,n})$ is connected if and only if $\text{PMCG}(S_{g,n}) \leq \langle\langle f \rangle\rangle$.

In [18], Putman provides a trick that simplifies the process of verifying whether a complex associated to a surface is connected, given below.

Proposition 3.3.6. Suppose G is a group and X is a graph on which G acts. Suppose G acts transitively on the vertices and that G is generated by the set $\{g_i\}$. If there is an edge e of X with the property that each g_i fixes at least one vertex of e , then X is connected.

We now cite some applications of the general well-suited curve criterion, due to Lanier and Margalit.

Proposition 3.3.7. Let $g \geq 0$, and let $f \in \text{MCG}(S_g)$. Suppose that there are nonseparating curves c and d in S_g so that $i(c, d) = 1$ and $i(f(c), d) = 0$. Then $\langle\langle f \rangle\rangle$ contains $[\text{MCG}(S_g), \text{MCG}(S_g)]$.

The following result allows us to upgrade the above proposition to conclude that f normally generates $\text{MCG}(S_g)$.

Theorem 3.3.8 (Harer). If $g \geq 3$, then $\text{PMCG}(S_{g,n})$ is perfect.

With some additional work, Margalit and Lanier also prove the following surprising results.

Theorem 3.3.9. For $g \geq 3$, every nontrivial periodic mapping class that is not a hyperelliptic involution normally generates $\text{MCG}(S_g)$.

Theorem 3.3.10. Let $g \geq 3$. For each $m \geq 0$ there are pseudo-Anosov mapping classes with arbitrarily large stretch factors whose normal closures in $\text{MCG}(S_g)$ are equal to $\text{MCG}(S_g)[m]$.

4. THE PURITY THEOREM

4.1. Big mapping class groups.

Recall that $\text{Homeo}(S, \partial S)$ is a topological group with the compact-open topology, hence $\text{MCG}(S)$ can be endowed with the quotient topology, which is again topological group. When S has finite type, $\text{MCG}(S)$ is countable; moreover, its quotient topology is discrete, so $\text{MCG}(S)$ can only be studied algebraically. On the other hand, when S has infinite type,

$\text{MCG}(S)$ is uncountable and its topology is no longer discrete (hence the name big mapping class groups), so the topological study of $\text{MCG}(S)$ is no longer trivial.

The fact that big mapping class groups are uncountable raises the following questions: Can we say anything about group homomorphisms $\text{MCG}(S) \rightarrow G$ when G is countable? How can we construct such homomorphisms?

4.2. Statement and applications.

We now introduce the motivating result of this survey: the Purity Theorem, which is proved in [8].

Theorem 4.2.1 (Calegari-Chen). Let S be a surface of finite type, and $K \subset S$ be a Cantor set. Let N be a normal subgroup of $\text{MCG}(S, K)$. Then either $N \subseteq \text{PMCG}(S, K)$ or $\ker \mathcal{F} \subseteq N$, where

$$\text{MCG}(S, K) \xrightarrow{\mathcal{F}} \text{MCG}(S)$$

is the forgetful map.

When seeking normal subgroups of $\text{MCG}(S, K)$ that satisfy the hypotheses of the Purity Theorem, one should observe that we can no longer obtain normal subgroups by pulling back from $\text{MCG}(S, K) \rightarrow \text{Homeo}(K)$, since $\text{Homeo}(K)$ is simple when K is a Cantor set [1].

The following corollary follows easily from the Purity Theorem.

Corollary 4.2.2. *A group homomorphism $\text{MCG}(S, K) \rightarrow G$ where G is countable factors through $\mathcal{F}$. In particular, if $\text{MCG}(S) = 1$, then $\text{MCG}(S, K)$ admits no nontrivial homomorphisms to a countable group.*

Proof. Let $\varphi : \text{MCG}(S, K) \rightarrow G$ be nontrivial. Then $G \cong \text{MCG}(S, K)/\ker \varphi$. Recall that $\text{MCG}(S, K)/\text{PMCG}(S, K) \cong \text{Homeo}(K)$ which is uncountable, hence $\ker \varphi \leq \text{PMCG}(S, K)$ implies that G is uncountable. By the Purity Theorem, $\ker \mathcal{F} \leq \ker \varphi$, so φ factors through $\mathcal{F}$. ■

The Purity Theorem also has interesting consequences regarding normal closures.

Proposition 4.2.3. Let $T \subset \text{MCG}(S, K)$ be a subset not contained in $\text{PMCG}(S, K)$. If $\langle\langle \mathcal{F}(T) \rangle\rangle = \text{MCG}(S)$, then $\langle\langle T \rangle\rangle = \text{MCG}(S, K)$.

Proof. Since $\langle\langle T \rangle\rangle$ is not contained in $\text{PMCG}(S, K)$, then $\ker \mathcal{F} \leq \langle\langle T \rangle\rangle$. But $\langle\langle \mathcal{F}(T) \rangle\rangle = \text{MCG}(S)$, hence $\langle\langle T \rangle\rangle = \mathcal{F}^{-1}\langle\langle \mathcal{F}(T) \rangle\rangle = \text{MCG}(S, K)$. ■

The Purity Theorem is a surprising result that relies heavily on the self-similarity of the Cantor set. As we will see in the next section, this self-similarity can be exploited to rewrite mapping classes in terms of infinite products and commutators, which would not be possible in other settings.

4.3. The proof.

We now sketch the proof of the Purity Theorem. Before doing so, we need to introduce some definitions. Throughout this section, S is a closed surface and $K \subset S$ is an embedded Cantor set. Furthermore, we assume that S has a marked point, which we denote by ∞ .

Definition 4.3.1. A disc $D \subset S - K$ is dividing if both the interior and exterior of D intersect K while its boundary does not (∞ must lie in the exterior of D).

Definition 4.3.2. We let N_{div} be the subgroup of $MCG(S, K)$ generated by all mapping classes supported in dividing discs.

As a first example of how the self-similarity of the Cantor set can be exploited algebraically in $MCG(S, K)$, we sketch a proof of the following Lemma, which is crucial in order to prove the Purity Theorem.

Lemma 4.3.3. *Let $g \in MCG(S, K)$ be non-pure. Then $N_{div} \leq \langle\langle g \rangle\rangle$*

Proof. Choose $x \in K$ such that $g(x) \neq x$, and let D be a dividing disc containing x such that $D \cap g(D) = \emptyset$. Let $\{D_n\}_{n \geq 0}$ be a sequence of disjoint dividing discs contained in D converging to x . There exists $h \in MCG(S, K)$ supported in D , preserving $K \cap D$ and fixing x , such that $h(D_n) = D_{n+1}$. Let $g(D) = E$ and $g(D_n) = E_n$, and let $b \in MCG(S, K)$ be such that $b|_D = g|_D$ and $b|_E = h \circ (g^{-1})|_E$.

Let $f \in MCG(S, K)$ be supported in a dividing disc, which up to conjugation we may assume to be D_0 . Then $x_f := \prod_{n \geq 0} h^n f^{-n}$ is a well-defined mapping class supported in D . Now, observe that $a_f := [x_f, g] = (x_f g x_f^{-1}) g^{-1} \in \langle\langle g \rangle\rangle$ (this is true for any $g \in MCG(S, K)$), and thus $a_f b a_f b^{-1} \in \langle\langle g \rangle\rangle$. By analyzing how a_f and $b a_f b^{-1}$ act on each E_n and D_n , we conclude that $a_f b a_f b^{-1}$ is supported in D_0 , on which it agrees with a_f , and thus with f . Therefore $a_f b a_f b^{-1} = f$. ■

By a simple diagram chase involving the Birman Exact Sequence, it suffices to show the result when S has a single puncture, which we denote ∞ ; the proof assumes that S is punctured at ∞ . A ray is the isotopy class of a path from a point K to ∞ that avoids K , and a lasso is a loop based at ∞ that is homotopically essential in S .

For simplicity, we let $\Gamma = MCG(S, K)$. We also introduce the following notation:

$$\begin{aligned} \Gamma^0 &= \ker \mathcal{F}, \\ \Gamma_r &= \text{Stab}(r), \\ \Gamma_{(r)} &= \{\text{Mapping classes that are Id on a nbhd of } r\}. \end{aligned}$$

The outline of the proof of the Purity Theorem is as follows. After proving that the normal closure of any $g \in \Gamma - \text{PMCG}(S, K)$ contains N_{div} , it suffices to show that $\Gamma^0 = N_{div}$. The latter is accomplished by the following sequence of lemmas, whose proof we omit.

Lemma 4.3.4. *For each finite filling collection of lassos L that are disjoint from r , the group $\Gamma_{L,(r)}$ is a subgroup of N_{div} , where $\Gamma_{L,(r)} \leq \Gamma_{(r)}$ consists of mapping classes that fix the isotopy class of each lasso in L .*

Lemma 4.3.5. *For any ray r , the group generated by all $\Gamma_{L,(r)}$ over all collections of lassos L filling and disjoint from r is equal to $\Gamma_{(r)} \cap \Gamma^0$.*

Lemma 4.3.6. *For any disjoint rays r and s , the group $\Gamma_r \cap \Gamma^0$ is generated by $\Gamma_{(r)} \cap \Gamma^0$ and $\Gamma_{(s)} \cap \Gamma_r \cap \Gamma^0$.*

Lemma 4.3.7. *Let r and s be disjoint rays. Then $\Gamma_r \cap \Gamma^0$ and $\Gamma_s \cap \Gamma^0$ generate Γ^0 .*

It is worth mentioning that the proof of Lemmas 4.3.5 and 4.3.6 rely heavily on the self-similarity of the Cantor set.

Having cited these Lemmas, we are now ready to prove the Purity Theorem.

Proof. Let $g \in \Gamma - \text{PMCG}(S, K)$. Then $N_{\text{div}} \leq \langle\langle g \rangle\rangle$ by Lemma 4.3.3. It remains to show that $N_{\text{div}} = \Gamma^0$, and it is clear that $N_{\text{div}} \leq \Gamma^0$. For the other inclusion, take any ray r and any collection L of filling lassos that are disjoint from r . Lemma 4.3.4 tells us that $\Gamma_{L,(r)} \leq N_{\text{div}}$. By varying L over all collections of filling lassos that are disjoint from r , Lemma 4.3.5 implies that $\Gamma_{(r)} \cap \Gamma^0 \leq N_{\text{div}}$ for any ray r . Let s and r be disjoint rays. Since $\Gamma_{(r)} \cap \Gamma^0$ and $\Gamma_{(s)} \cap \Gamma_r \cap \Gamma^0$ are contained in N_{div} , Lemma 4.3.6 implies that $\Gamma_r \cap \Gamma^0 \leq N_{\text{div}}$, and Lemma 4.3.7 implies that $N_{\text{div}} \leq \Gamma^0$. ■

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